

A Density-One Set with Distinct Consecutive-Block Products

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Abstract

We construct a set $A \subset \mathbb{N}$ of natural density one whose increasing enumeration $d_1 < d_2 < \dots$ has distinct products on finite consecutive blocks. The construction is a modified greedy algorithm with a deterministic density-zero deletion set. This set turns semiprime candidates supplied by the Matomäki–Teräväinen theorem into private barriers. After density-zero exceptional families are removed, the barriers localize the chosen nonempty old-new collision witness to a polylogarithmic window. The localized identities yield many large-prime congruences of shifted-CRT type. A low-degree auxiliary polynomial then vanishes at the global collision point; nonlinear components are counted using the Castryck–Cluckers–Dittmann–Nguyen affine-curve bound, while line components are counted directly from the curvature of q^θ . This gives $o(X)$ remaining collision endpoints in every dyadic shell $(X, 2X]$, and hence a density-one accepted set with distinct consecutive-block products.

1 Introduction

For an increasing sequence $d_1 < d_2 < \dots$ of positive integers, write

$$P_d(u, v) = \prod_{i=u}^v d_i \quad (1 \leq u \leq v).$$

The sequence has *distinct consecutive-block products* if

$$P_d(u, v) = P_d(u', v') \implies (u, v) = (u', v')$$

for all finite index intervals. Erdős asked whether this is compatible with natural density one; this is Problem 421 in the Erdős problem collection [2], originating from Erdős and Graham’s problem list [1].

Theorem 1.1. *There exists a strictly increasing sequence*

$$1 \leq d_1 < d_2 < \cdots$$

whose underlying set has natural density 1, and for which the products $\prod_{i=u}^v d_i$ are all distinct as (u, v) ranges over finite consecutive intervals.

The proof is self-contained modulo the external results stated in Section 2. We use a modified greedy construction. First build a deterministic set D of density zero. Integers in D are forcibly rejected, 2 is accepted, and every later integer outside D is accepted exactly when adjoining it creates no new equality between distinct consecutive-block products. The greedy rule then guarantees the required uniqueness; the problem is to arrange that the forced and greedy rejections still have density zero.

The main obstruction is the following. If a nonforced endpoint n is rejected, then there is an old-new witness

$$P = nS,$$

where P is an earlier consecutive-block product and S is a final suffix product of the accepted prefix. A direct block-product count handles $S = 1$. For $S > 1$, the old block P may be long and far from the suffix; the role of D is to force almost every such witness into a short window where it can be counted.

The set D is designed to produce a *barrier system*. In most polylogarithmic blocks one selects a semiprime $b = p_1 p_2$ with $(\log b)^{1.08} < p_1 \leq (\log b)^{1.1}$ and with a globally private large prime label p_2 . All other multiples of p_2 , up to a long finite horizon, are deleted. This has two consequences. First, the selected barriers survive the greedy scan, since any new product containing such a barrier has a private prime divisor which no old product can have. Second, any surviving collision whose old block crosses a barrier is impossible unless the endpoint lies beyond that barrier's horizon. The finite-horizon exceptions form a density-zero smooth family.

After this reduction, every remaining nonempty collision in a shell $(X, 2X]$ has a short, controlled witness of the form

$$\prod_{i=1}^{\ell} (x - a_i) = \prod_{j=1}^b (q + c_j), \quad 0 \leq a_i, c_j \leq (\log X)^{O(1)},$$

with $x < q < n \leq 2X$, q a previous selected barrier, and

$$x = q^{b/\ell} + O((\log X)^B), \quad (\log X)^{-B} \leq 1 - \frac{b}{\ell} \ll \frac{\log \log X}{\log X}.$$

Large prime factors of the left side must occur among the terminal factors on the right. This supplies many congruences

$$p_i \mid x - a_i, \quad p_i \mid q - x + a_i + c_i,$$

with distinct $p_i > \exp(\log X / (\log \log X)^2)$. A low-degree auxiliary polynomial vanishing on the local points $(a_i, a_i + c_i)$ must therefore vanish at the global point $(x, x - q)$. Nonlinear components are counted by the affine-curve determinant-method theorem of Castryck–Cluckers–Dittmann–Nguyen; line components are counted directly from the strict curvature of q^θ for $\theta < 1$.

The paper is organized as follows. Section 2 fixes the external inputs and notation. Section 3 records the formal greedy reduction and disposes of empty-suffix collisions. Section 4 constructs the barrier system. Sections 5 and 6 turn the barrier information into localized product identities and then into terminal large-prime support. Section 7 proves the shifted-CRT counting proposition. Section 8 applies it to all remaining nonempty rejections, and Section 9 completes the density argument.

2 External Inputs and Notation

All logarithms are natural, except $\log_2$. Constants implicit in $O(\cdot)$, $o(\cdot)$, and $\ll$ may depend on the fixed constants chosen in the proof, but not on the dyadic parameter X . A *dyadic shell* means $(X, 2X]$, with X a sufficiently large power of two. For an integer $m \geq 2$, let $P^+(m)$ be its largest prime factor. Let

$$\Psi(X, Y) = \#\{m \leq X : P^+(m) \leq Y\}.$$

External input 2.1 (Matomäki–Teräväinen E_2 theorem [3]). *There are constants $c_0, \delta > 0$ such that, for all sufficiently large X , all but $O(X/(\log X)^\delta)$ integer starts $2 \leq y \leq X$ satisfy*

$$\#\left\{p_1 p_2 \in (y, y + (\log y)^{2.1}] : \begin{array}{l} p_1, p_2 \text{ prime,} \\ (\log y)^{1.09} < p_1 \leq (\log y)^{1.1} \end{array} \right\} \geq c_0 (\log y)^{1.1}.$$

External input 2.2 (de Bruijn–Dickman smooth-number estimate [4, 5]). *Uniformly for $u = u(N) \rightarrow \infty$ with $u = O(\sqrt{\log \log N})$,*

$$\Psi(N, N^{1/u}) \leq N \exp\left(-(1 + o(1))u \log u\right).$$

External input 2.3 (CCDN affine-curve bound [6]). *For every fixed ambient dimension $n > 1$, there is a constant C_n such that, for every $B \geq 1$, every integral affine curve $V \subset \mathbb{A}_{\mathbb{Q}}^n$ of positive degree d satisfies*

$$\#\{z \in V(\mathbb{Z}) : \text{ht}(z) \leq B\} \leq C_n d^3 B^{1/d} (\log B + d).$$

Here $\text{ht}(z) = \max_i |z_i|$ for $z = (z_1, \dots, z_n) \in \mathbb{Z}^n$. The constant C_n is independent of d , B , and the defining equations of V ; the displayed factor is the only degree-dependence used below.

External input 2.4 (Mignotte factor inequality [7]). *If $f, g \in \mathbb{Z}[z]$ are nonzero, $\deg g \leq d$, and g is a multiple of f , then*

$$\|g\|_2 \geq 2^{-d} \|f\|_{\infty}.$$

We also use the following standard elementary facts: Bertrand's postulate, Chebyshev upper and lower prime-counting bounds, Gauss's lemma for primitive factors over $\mathbb{Q}$ and $\mathbb{Z}$, the usual compatibility of primitive homogenization and dehomogenization for line factors, Galois conjugacy of absolute factors of a squarefree $\mathbb{Q}$ -irreducible plane curve, and Bezout's $d_1 d_2$ bound for two coprime plane curves of degrees d_1, d_2 .

3 The Modified Greedy Reduction

This section separates the formal consequences of the greedy rule from the number-theoretic construction of D . Once D has density zero, the only greedy rejections that require serious work are the nonempty old-new collisions defined below.

Definition 3.1 (Modified greedy sequence). Fix a deterministic set $D \subset \mathbb{N}$ with $1 \in D$, $2 \notin D$, and $|D \cap [1, N]| = o(N)$. The modified greedy construction rejects every $n \in D$, accepts 2, and for each $n \geq 3$ with $n \notin D$, accepts n if and only if adjoining n to the accepted prefix creates no equality between two distinct finite consecutive-block products.

Lemma 3.2 (Greedy uniqueness). *Every finite accepted prefix produced by Definition 3.1 has distinct consecutive-block products.*

Proof. The initial accepted prefix is $\{2\}$. A rejection leaves the accepted prefix unchanged. An acceptance occurs only after the enlarged prefix has been tested to have no equality between two distinct consecutive-block products. Induction over the scan proves the claim. $\square$

Lemma 3.3 (Old-new witness split). *Let $n \geq 3$ be rejected by the modified greedy construction and suppose $n \notin D$. If $d_1 < \dots < d_m$ is the accepted prefix before testing n , then there is an old consecutive accepted block product P and a suffix product S of the old prefix, possibly $S = 1$, such that*

$$P = nS.$$

Proof. The old prefix has distinct block products by Lemma 3.2. Since $n \notin D$, rejection means that adding n creates a collision. Old-old collisions are impossible. Two new block products both contain n ; after cancelling n , one obtains equality of two final suffix products of the old prefix. Distinct final suffixes are strictly ordered by inclusion, and extending a suffix multiplies its product by an accepted integer at least 2. Hence two new products cannot collide. The collision is therefore between an old block and a new block, and every new block is n times an old suffix product. $\square$

Definition 3.4. A nonforced rejection $n \notin D$ is an *empty-suffix rejection* if it has some witness in Lemma 3.3 with $S = 1$. Otherwise we choose, once and for all, the lexicographically first old-new witness in accepted-index coordinates, first by old-block endpoints and then by suffix start, and call n a *nonempty rejection*. For nonempty rejections we cancel the maximal common contiguous overlap between the old block and the final suffix. The old block is not contained in that suffix, and if the cancellation exhausted the suffix then n would itself be an old block product. Thus the remaining suffix is nonempty, and the old block lies strictly before it.

Lemma 3.5 (Sparse old products). *For any increasing positive-integer sequence $d_1 < d_2 < \dots$ with $d_1 \geq 2$, the number of values at most N which occur as $P_d(u, v)$ with $v - u + 1 \geq 2$ is $o(N)$.*

Proof. Since $d_i \geq i + 1$, a length- ℓ block starting at index i has product at least $(i + 1)^\ell$. For fixed $\ell \geq 2$, at most $N^{1/\ell}$ length- ℓ block products can be $\leq N$. No length $\ell > \log_2 N$ contributes. Thus the number of such values is at most

$$\sum_{2 \leq \ell \leq \log_2 N} N^{1/\ell} = O(N^{1/2} \log N) = o(N).$$

$\square$

Corollary 3.6. *The empty-suffix rejections of the modified greedy construction have density zero.*

Proof. An empty-suffix rejection n is an old block product. The old block has length at least two, since length one would make n an already accepted old term. Apply Lemma 3.5 to the accepted sequence. $\square$

Lemma 3.7 (Elementary smooth decay). *Let $E > 0$, $\rho > 1$, and $M > 0$ be fixed. Then*

$$\Psi(X, (\log X)^E) = o(X)$$

and, with $Y_X = \exp(\log X / (\log \log X)^\rho)$,

$$\Psi(2X, Y_X) = o\left(\frac{X}{(\log X)^M}\right).$$

Proof. For $1/2 \leq \sigma < 1$, Rankin's trick gives

$$\Psi(X, Y) \leq X^\sigma \prod_{p \leq Y} (1 - p^{-\sigma})^{-1}.$$

Write $\sigma = 1 - \eta$, $0 < \eta \leq 1/2$. Chebyshev's bound and partial summation give

$$\log \prod_{p \leq Y} (1 - p^{-\sigma})^{-1} \ll \log \log(3Y) + \frac{Y^\eta}{\eta \log Y}.$$

Indeed, this follows from

$$\sum_{p \leq Y} p^{-1+\eta} \leq \pi(Y)Y^{-1+\eta} + (1-\eta) \int_2^Y \pi(t)t^{-2+\eta} dt \ll \frac{Y^\eta}{\log Y} + \int_2^Y \frac{t^{-1+\eta}}{\log t} dt,$$

and splitting the last integral at $\log t = 1/\eta$, which gives $O(\log \log(3Y))$ below the split and $O(Y^\eta/(\eta \log Y))$ above it.

For $Y = (\log X)^E$, choose fixed $\sigma \in (0, 1)$ with $E(1 - \sigma) < 1$, obtaining $X^\sigma \exp(o(\log X)) = o(X)$. For $Y_X = \exp(L/u)$, $L = \log X$, $u = (\log \log X)^\rho$, take $\eta = (\log u)/(2 \log Y_X)$. Then $Y_X^\eta = \sqrt{u}$ and $\eta L = \frac{1}{2}u \log u$, while the Euler-product exponent above is

$$O(\log \log X + \sqrt{u}/\log u) = o(u \log u).$$

Therefore

$$\Psi(2X, Y_X) \leq 2X \exp(-\Omega(u \log u)),$$

which beats every fixed power of $\log X$ because $\rho > 1$. $\square$

Lemma 3.8 (Actual blocks and final suffixes). *For any strictly increasing accepted set:*

- (a) for fixed accepted endpoint x and fixed length ℓ , there is at most one actual consecutive accepted block of length ℓ ending at x ;
- (b) for fixed $r \geq 0$, the function

$$F_r(n) = n \prod_{\substack{a < n \\ a \text{ accepted} \\ a \text{ among last } r}} a$$

is strictly increasing on candidates n for which the last r accepted predecessors exist.

Proof. Part (a) is immediate from strict increase. If $n_1 < n_2$, then the ordered r -tuple of last accepted predecessors before n_1 is coordinatewise $\leq$ the corresponding tuple before n_2 . The predecessor product is therefore nondecreasing, and multiplication by $n_1 < n_2$ gives $F_r(n_1) < F_r(n_2)$. $\square$

4 Private Barriers

We now construct the deterministic set D . This is the only point at which the barriers are built; later sections use only the three properties stated in Proposition 4.1. Put

$$\beta = 6, \quad L_X = \left\lfloor (\log X)^\beta \right\rfloor, \quad L_0(X) = (\log(3X))^{2.1}.$$

For each sufficiently large dyadic shell $(X, 2X]$, partition the shell into full consecutive intervals of length L_X ; boundary remainders are put in D . In a full block $I = [U, U + L_X)$, an integer start y is *admissible* if $U \leq y$ and $y + L_0(X) < U + L_X$.

Proposition 4.1 (Finite-horizon private barriers). *There is a deterministic set $D \subset \mathbb{N}$ of density zero and a set of selected semiprime barriers $b = p_1 p_2$ such that:*

- (i) every selected barrier has

$$(\log b)^{1.08} < p_1 \leq (\log b)^{1.1},$$

and its large prime p_2 is a globally unique private label;

- (ii) every selected barrier lies outside D and is accepted by the modified greedy construction;

(iii) for every sufficiently large N , put

$$F_Y = Y^{\lfloor \sqrt{\log \log Y} \rfloor},$$

and let $X_0 = X_0(N)$ be the dyadic scale with $F_{X_0} < N \leq F_{2X_0}$. Outside D , a finite initial segment, and the $o(N)$ inverse-horizon smooth family whose chosen witness has old-left maximum $x \leq 5X_0$, every nonforced nonempty rejection $n \leq N$, with its chosen overlap-cancelled witness, has a previous selected barrier $q_B(n) < n$ such that

$$n - q_B(n) = O((\log n)^\beta),$$

the final suffix of the witness starts after $q_B(n)$, and the old-left block in the witness has value span $O((\log n)^\beta)$.

Only the properties listed in Proposition 4.1 are used in the later localization and counting arguments.

Proof. We start above a large fixed dyadic shell, put $1 \in D$, keep $2 \notin D$, and absorb all finite initial choices into $O(1)$.

Good blocks and label supply. Call a full L_X -block bad if fewer than half of its admissible starts are good in the sense of Input 2.1. Since the admissible-start sets of distinct full blocks are disjoint and each full block has $(1 + o(1))L_X$ admissible starts, Input 2.1, applied up to $2X + O(L_X)$, implies that bad blocks in $(X, 2X]$ have total length $O(X/(\log X)^\delta)$. Put all bad blocks in D . For every admissible start in the shell, $y < 3X$ for large X , so $(\log y)^{2.1} \leq L_0(X)$; hence the MT interval attached to y remains inside its long block.

In any remaining block, the good starts contribute $\gg L_X(\log X)^{1.1}$ start-semiprime incidences. A fixed semiprime b can be counted only for starts y with $b - L_0(X) < y < b$, hence for at most $L_0(X) + O(1)$ starts. Thus the block contains at least

$$\gg \frac{L_X(\log X)^{1.1}}{L_0(X)} = (\log X)^{\beta-1-o(1)}$$

distinct semiprimes. A fixed large prime label p_2 gives at most one such semiprime in one long block: two products $p_1 p_2, p'_1 p_2$ differ by at least p_2 , while for these semiprimes $p_2 = b/p_1 \gg X/(\log X)^{1.1} \gg L_X$. Hence each good block has $(\log X)^{\beta-1-o(1)}$ distinct candidate labels. For every counted $b = p_1 p_2$, since $b \in (y, y + L_0(X)]$, we have $\log b = \log y + O(L_0(X)/X) = (1 + o(1)) \log X$. The upper bound below is immediate, and the lower bound follows from the exponent gap $1.09 > 1.08$; after increasing the initial cutoff,

$$(\log b)^{1.08} < p_1 \leq (\log b)^{1.1}.$$

Greedy block-label matching. Order good blocks by left endpoint and candidate labels in each block by increasing p_2 . Scan blocks in order and match a block to the first candidate label which was not selected in an earlier shell and has not already been matched in the current shell. If no such label exists, leave the block unmatched.

A label selected in an earlier shell $(Y, 2Y]$ can also be a candidate in $(X, 2X]$ only for polylogarithmically comparable Y . Indeed,

$$\frac{X}{(\log X)^{1.1}} \ll p_2 \ll \frac{X}{(\log X)^{1.08}}, \quad \frac{Y}{(\log Y)^{1.1}} \ll p_2 \ll \frac{Y}{(\log Y)^{1.08}},$$

first gives $Y = X^{1+o(1)}$, and then the safer window

$$cX(\log X)^{-0.05} \leq Y \leq CX(\log X)^{0.05}.$$

Thus the number of earlier selected labels relevant to the current shell is

$$O\left(\frac{X}{L_X}(\log X)^{0.06}\right),$$

where the factor $(\log X)^{0.01}$ absorbs the $O(\log \log X)$ dyadic shells in the comparable range. Current-shell matched labels are absorbed in the same bound. Also a fixed label is adjacent to at most $(\log X)^{1.1+o(1)}$ current blocks: p_1 has only that many possible prime values, and each product $p_1 p_2$ lies in $O(1)$ long blocks. If U good blocks remain unmatched, then every candidate label of each unmatched block is either an earlier selected label or a label already matched in the current shell. Counting these incidences gives

$$U(\log X)^{\beta-1-o(1)} \ll \frac{X}{L_X}(\log X)^{0.06}(\log X)^{1.1+o(1)}.$$

Hence

$$U = O\left(\frac{X}{L_X}(\log X)^{-(\beta-2.16)+o(1)}\right) = o(X/L_X).$$

Put all unmatched good blocks in D . In each matched block select the matched semiprime as its barrier.

Prefix deletions. After every maximal deleted run of long blocks, and after each shell boundary, look at the first subsequent matched block and delete its nonbarrier prefix before the selected barrier. The number of bad blocks is $O(X/(L_X(\log X)^\delta))$, and the number of unmatched blocks is

$$O\left(\frac{X}{L_X}(\log X)^{-(\beta-2.16)+o(1)}\right).$$

Multiplying by the prefix length L_X , the block and prefix deletions in $(X, 2X]$ have size

$$O(X/(\log X)^\delta) + O\left(X(\log X)^{-(\beta-2.16)+o(1)}\right) + O(L_X) = o(X).$$

Finite-horizon multiples. For a selected barrier in shell X , let

$$C_X = \left\lfloor \sqrt{\log \log X} \right\rfloor, \quad F_X = X^{C_X}.$$

For its private label p_2 , put into D every multiple $m \neq b$ of p_2 with $m \leq F_X$. These horizon deletions have density zero. A shell $(X, 2X]$ contributes at most

$$O\left(\frac{\min(N, F_X)}{(\log X)^{\beta-1.1}}\right)$$

to $D \cap [1, N]$, since there are $O(X/L_X)$ selected labels and each label is $\gg X/(\log X)^{1.1}$. Put $s = \beta - 1.1 > 1$.

For old shells with $F_X < N$, the dyadic sum is dominated by the last such shell because C_X is nondecreasing and

$$F_{2X}/F_X \geq 2^{\frac{1}{2}\sqrt{\log \log X}}$$

eventually. This gives $O(N/(\log X_0)^s)$, where X_0 is the largest dyadic shell with $F_{X_0} < N$. Recent shells $X_0 < X \leq N$ contribute

$$N \sum_{X_0 < X \leq N} \frac{1}{(\log X)^s} = O\left(\frac{N}{(\log X_0)^{s-1}}\right) = o(N),$$

because $F_{X_0} < N \leq F_{2X_0}$ implies $\log X_0 \geq (1 + o(1)) \log N / \sqrt{\log \log N}$. Future shells $X > N$ can affect $[1, N]$ only if $X/(\log X)^{1.1} \leq N$, hence $X \leq N(\log N)^2$ for large N ; their total contribution is

$$O\left(\frac{N \log \log N}{(\log N)^s}\right) = o(N).$$

Finally, the other deterministic deletions have dyadic total $o(N)$. The boundary remainders contribute $O((\log N)^{\beta+1})$, and the MT-bad blocks contribute $O(N/(\log N)^\delta)$. The unmatched-block and prefix contribution is $O(X(\log X)^{-(\beta-2.16)+o(1)})$ in a shell; since $\beta = 6$, its dyadic total is also $o(N)$. This proves $|D \cap [1, N]| = o(N)$.

Barrier survival and acceptance. A selected barrier is in a matched block, is not in a deleted prefix, and is protected from its own label deletion. No other

selected label deletes it. Indeed, if a selected label p from shell Y divides $b = p_1 p_2$ and $b \leq F_Y$, then $p = p_1$ or $p = p_2$. The case $p = p_2$ contradicts global label uniqueness. The case $p = p_1$ is impossible for large Y , because

$$p \geq \frac{Y}{(\log Y)^{1.1}}, \quad p_1 \leq (\log b)^{1.1} \leq (\log F_Y)^{1.1} = (C_Y \log Y)^{1.1}$$

and the last quantity is

$$o\left(\frac{Y}{(\log Y)^{1.1}}\right).$$

When b is scanned, every earlier multiple of p_2 has been deleted, since $b \leq 2X < F_X$. Thus no old accepted product is divisible by p_2 : if p_2 divided such a product, it would divide one old accepted factor, which would be an earlier deleted multiple of p_2 . Every new product containing b , on the other hand, is divisible by p_2 . Hence no old-new collision can reject b . Old-old collisions are excluded by Lemma 3.2. Finally, two new products containing b would, after cancelling b , give equality between two final suffix products of the old accepted prefix; distinct final suffixes are strictly ordered by inclusion and each proper extension multiplies by an accepted term at least 2. Hence no new-new collision can reject b . Therefore each selected barrier is accepted. *Blocking.* Consider a surviving overlap-cancelled witness

$$\prod_{\text{old left}} d_i = n \prod_{\text{final suffix}} d_i$$

with the old-left block strictly before the final suffix. Suppose the old left block contains a selected barrier $b = p_1 p_2$ from shell X , and $n \leq N \leq F_X$. Then p_2 divides the left product, hence divides the right product. The corresponding terminal factor is either n or a suffix factor. It is not b , because the right side lies strictly to the right of the old block. It is not any selected barrier $c = r_1 r_2 \leq F_X$: if $p_2 = r_2$, global label uniqueness is violated; if $p_2 = r_1$, then

$$p_2 \leq (\log c)^{1.1} \leq (\log F_X)^{1.1} = o(X/(\log X)^{1.1}),$$

contradicting $p_2 \gg X/(\log X)^{1.1}$. Thus the terminal factor is a nonbarrier multiple of p_2 below F_X , hence belongs to D , which is impossible for either n or an accepted suffix factor.

Previous barriers and localization. Every sufficiently large nondeleted integer y which is not itself a selected barrier has a previous selected barrier $q_B(y) < y$ with

$$y - q_B(y) = O((\log y)^\beta).$$

If y lies after the selected barrier in its matched block, use that barrier. If y lies before it, then y is in the nonbarrier prefix of that matched block. Since $y \notin D$, this block cannot be the first matched block after a deleted run or after the shell boundary, because those prefixes were deleted. The preceding full block is therefore undeleted and matched, and its barrier is below y within two L_X -blocks.

Let n be a surviving nonempty rejection. It is not itself a selected barrier, since selected barriers are accepted. The final suffix in its chosen witness contains no selected barrier: if it contained $b = p_1 p_2$, then every old-left factor would be $< b \leq 2X_b < F_{X_b}$, where X_b is the shell of b . An old-left factor divisible by p_2 could not be a selected barrier, by global label uniqueness and the small-factor bound already used above; it would therefore be a deleted nonbarrier multiple of p_2 . Thus the left product would not be divisible by p_2 , while the right product would be, a contradiction. If the final suffix began at or before $q_B(n)$, it would contain the accepted selected barrier $q_B(n)$. Consequently the suffix starts after $q_B(n)$, and its length is $O((\log n)^\beta)$.

Finally choose, for large N , the dyadic $X_0 = X_0(N)$ with

$$F_{X_0} < N \leq F_{2X_0}.$$

If the old-left maximum x in the chosen witness satisfies $x \leq 5X_0$, then n is $5X_0$ -smooth, because n divides the old-left product. Moreover

$$\frac{\log N}{\log(5X_0)} = (1 + o(1)) \sqrt{\log \log N}.$$

Input 2.2 shows that these endpoints are $o(N)$. For all remaining endpoints, $x > 5X_0$. If x is a selected barrier, its horizon is at least N : the dyadic scale of x is above $2X_0$, and $X \mapsto F_X = X^{\lfloor \sqrt{\log \log X} \rfloor}$ is eventually increasing on dyadic scales, so this horizon is at least $F_{2X_0} \geq N$. The blocking argument forbids survival. If x is not a selected barrier, the previous barrier below x lies within $O((\log x)^\beta)$ and, for large N , above $2X_0$, since $x \leq N \leq F_{2X_0}$ makes every fixed power of $\log x$ equal to $o(X_0)$. The same monotonicity gives horizon at least N . The old-left block cannot contain that previous barrier, so it must start after it. Thus the old-left span is $O((\log n)^\beta)$. $\square$

5 Localized Nonempty Witnesses

Fix the deletion set and selected barriers from Proposition 4.1. For a nonempty rejection n , use the deterministic overlap-cancelled witness from Section 3. The purpose of this section is to convert the qualitative barrier information

into a numerical product identity with all local shifts bounded by a fixed power of $\log X$.

Proposition 5.1 (Localized witness). *There is a fixed exponent $B > 20$ such that, in every sufficiently large dyadic shell $X < n \leq 2X$, all but $o(X)$ nonempty rejection endpoints have a chosen witness*

$$\prod_{i=1}^{\ell} (x - a_i) = \prod_{j=1}^b (q + c_j), \quad q = q_B(n), \quad n = q + h,$$

with the following properties:

$$\begin{aligned} 2 \leq h &\leq (\log X)^7, & 0 \leq a_i &< T_X, & 1 \leq c_j &\leq h, & T_X &= (\log X)^B, \\ b &= r + 1, & \ell &= b + s, & s &\geq 1, & r &\leq (\log X)^7, \\ \ell &\leq \lceil ((\log X)^7 + 1) \log_2(2X) \rceil + 2, \\ b &\geq c \frac{\log X}{\log \log X} \quad \text{for a fixed } c > 0, \\ x &= q^{b/\ell} + O(T_X) \quad \text{with an absolute implicit constant,} \\ (\log X)^{-B} &\leq 1 - \frac{b}{\ell} \leq B \frac{\log \log X}{\log X}, \\ x &< q < 2X. \end{aligned}$$

The left factors $x - a_i$ are distinct accepted terms, the terminal values $q + c_j$ are distinct, and they are exactly $n = q + h$ together with the actual final length- r suffix before n .

Proof. The witness split gives an equality between an old block and n times a final suffix. The old block cannot be wholly contained in that suffix, since then its product would be at most the suffix product, whereas the right side is n times the suffix product with $n > 1$. If it overlaps the suffix, cancel the common contiguous overlap. Since the suffix is final, these are the only possibilities besides the old block already lying strictly before the suffix. If cancellation exhausted the suffix, then n would be an old block product and hence an empty-suffix rejection. Thus the chosen witness has the form

$$\prod_{t=u}^v d_t = n \prod_{t=c}^m d_t, \quad v < c \leq m.$$

Let $x = d_v$, let $w = d_u$, and let $r = m - c + 1$. Proposition 4.1, applied with ambient bound $2X$, removes $o(X)$ deterministic, inverse-horizon smooth, and

finite-initial endpoints and gives

$$\begin{aligned} 2 \leq h = n - q_B(n) &\leq (\log X)^7, \\ r &\leq (\log X)^7, \\ x - w &\leq (\log X)^7, \\ x &> (\log X)^7. \end{aligned}$$

After this removal, the remaining endpoints have $x > 5X_0(2X)$. This inverse-horizon scale satisfies $\log X_0(2X) = (1+o(1)) \log X / \sqrt{\log \log X}$, so $X_0(2X) > (\log X)^C$ for every fixed C , once X is large; in particular $x > (\log X)^7$. The lower bound $h \geq 2$ follows because the nonempty suffix starts after $q_B(n)$ and contains an accepted term $< n$.

The terminal side has $r+1$ factors, each at most $2X$, while every old-left factor is an accepted integer at least 2. Hence

$$2^\ell \leq (2X)^{r+1},$$

which gives the stated upper bound for ℓ . Also $\ell \geq r+2$: all terminal factors are larger than all old-left factors, so if $\ell \leq r+1$, the terminal product would strictly exceed the old product.

Two elementary tails are negligible. First suppose

$$\frac{\ell}{r+1} \geq 1 + 60 \frac{\log \log X}{\log X}.$$

Since the old-left span is at most $(\log X)^7$,

$$(x - (\log X)^7)^\ell \leq (2X)^{r+1},$$

so $x \leq (\log X)^7 + X/(\log X)^{40} \leq X/(\log X)^{39}$ for large X . For fixed x, ℓ, r , the old-left block and its product are fixed, and the right side

$$n \cdot (\text{last } r \text{ accepted predecessors before } n)$$

is strictly increasing by Lemma 3.8. Thus there is at most one endpoint for each fixed x, ℓ, r . Here x has $O(X/(\log X)^{39})$ possible values, while ℓ and r have only fixed-polylogarithmically many possible values; the total is therefore $O(X/(\log X)^{20}) = o(X)$. Second, if

$$r \leq \left\lfloor \frac{\log X}{60 \log \log X} \right\rfloor,$$

then the same argument gives

$$x \leq (\log X)^7 + (2X)^{(r+1)/(r+2)} \leq \frac{3X}{(\log X)^{50}},$$

and the same fixed-product and monotonicity argument, now with only polylogarithmically many (ℓ, r) and $O(X/(\log X)^{50})$ possible values of x , gives $o(X)$ endpoints.

For the remaining endpoints, with $b = r + 1$,

$$b \geq c \frac{\log X}{\log \log X}, \quad \ell = b + s, \quad 1 \leq s < 60b \frac{\log \log X}{\log X}.$$

Choose fixed $B > 20$ so large that $T_X = (\log X)^B$ dominates all fixed polylogarithmic constants in the preceding estimates. The witness becomes

$$\prod_{i=1}^{\ell} (x - a_i) = \prod_{j=1}^b (q + c_j),$$

where $0 \leq a_i < T_X$, $1 \leq c_j \leq h$, and $n = q + h$ is included among the terminal values.

The terminal product lies between q^b and $(q + h)^b$, and the old product lies between $(x - (\log X)^7)^\ell$ and x^ℓ . Thus

$$q^{b/\ell} \leq x \leq (\log X)^7 + (q + h)^{b/\ell}.$$

Since $0 < b/\ell < 1$, the function $t^{b/\ell}$ is increasing and concave on $[q, q + h]$, with derivative at most 1. Hence $(q + h)^{b/\ell} - q^{b/\ell} \leq h$, and therefore

$$0 \leq x - q^{b/\ell} \leq (\log X)^7 + h \leq T_X$$

for large X , after enlarging B if necessary. The lower bound

$$1 - \frac{b}{\ell} \geq (\log X)^{-B}$$

follows from $s \geq 1$ and $\ell \leq (\log X)^B$; the upper bound follows from $s < 60b \log \log X / \log X$, after enlarging B .

Finally let $\theta = b/\ell$ and $\Delta = 1 - \theta \geq (\log X)^{-B}$. Since $q \geq X - (\log X)^7$, the function $\Delta \mapsto 1 - e^{-\Delta \log q}$ is increasing, and $(\log X)^{-B} \log q = o(1)$, we have

$$q - q^\theta = q(1 - e^{-\Delta \log q}) \gg \frac{X \log X}{(\log X)^B} \gg T_X.$$

Together with $0 \leq x - q^\theta \leq T_X$, this gives $x < q$. The inequality $q < 2X$ is immediate from $q < n \leq 2X$. $\square$

6 Terminal Support

The next two lemmas turn the localized product identity into many independent large-prime congruences.

Lemma 6.1 (Smooth accepted left factors). *Let*

$$Y_X = \exp\left(\frac{\log X}{(\log \log X)^2}\right).$$

Among the localized endpoints of Proposition 5.1, those for which some accepted left factor $x - a_i$ has $P^+(x - a_i) \leq Y_X$ form an $o(X)$ family in the shell $(X, 2X]$.

Proof. By Lemma 3.7, the number of Y_X -smooth integers $\leq 2X$ is

$$o\left(\frac{X}{(\log X)^M}\right)$$

for every fixed M . Given such an integer $m = x - a_i$, there are at most T_X choices for x . For fixed x and ℓ there is at most one actual left block, and for each fixed suffix length r at most one endpoint. Indeed, the fixed left block has a fixed product P , while any endpoint counted with that block and this r must satisfy

$$P = n \prod_{\substack{a < n \\ a \text{ accepted} \\ a \text{ among last } r}} a,$$

and the right side is strictly increasing in n by Lemma 3.8. These at most T_X choices of x already account for the possible position of the smooth factor inside the left block. The remaining choices of ℓ and r are at most $O((\log X)^8)$ and $O((\log X)^7)$, respectively. Thus the endpoint count is

$$o\left(\frac{X}{(\log X)^M}\right) O((\log X)^{B+15}),$$

and taking M larger than this fixed loss gives $o(X)$. □

Lemma 6.2 (Terminal-support extraction). *Consider a localized endpoint from Proposition 5.1, not counted in Lemma 6.1, and take X large enough that $(\log X)^7 < T_X$ and $T_X + (\log X)^7 < Y_X$. For every left factor choose*

$$p_i = P^+(x - a_i) > Y_X.$$

Put

$$\Lambda_X = \left\lceil \frac{\log(3X)}{\log Y_X} \right\rceil + 2.$$

Then the endpoint supplies at least $\lfloor \ell/\Lambda_X^2 \rfloor$ distinct local slab points (a_i, r_i) , $r_i = a_i + c_i$, with distinct r_i 's and carrying distinct prime labels $p_i > Y_X$ such that

$$p_i \mid x - a_i, \quad p_i \mid D_0 + r_i, \quad D_0 = q - x > 0.$$

Proof. Since p_i divides the left product, it divides one terminal value $q + c_i$, $1 \leq c_i \leq h$. This terminal value is unique, because two terminal values differ by less than $h < Y_X < p_i$. A label p_i appears on at most one left factor and at most one terminal value, since left factors differ by less than $T_X < Y_X$ and terminal values differ by at most $h < Y_X$.

A fixed terminal value $\leq n \leq 2X < 3X$ is incident to at most Λ_X chosen labels. Indeed, the chosen labels incident to it are distinct primes, and it cannot have more than $\log(3X)/\log Y_X + 1$ distinct prime factors exceeding Y_X . Therefore the incidence graph from ℓ left vertices to terminal vertices contains a matching of size at least $\lfloor \ell/\Lambda_X \rfloor$: the ℓ left edges occupy at least $\lfloor \ell/\Lambda_X \rfloor$ terminal vertices, and choosing one incident edge over each occupied terminal gives a matching, since each left vertex has degree one. The matched labels are distinct.

For a matched edge, $p_i \mid x - a_i$ and $p_i \mid q + c_i$, so

$$p_i \mid (q + c_i) - (x - a_i) = D_0 + a_i + c_i.$$

Set $r_i = a_i + c_i$. For a fixed value ρ , all matched labels with $r_i = \rho$ divide the positive integer $D_0 + \rho$, where positivity follows from $D_0 > 0$ and $\rho = a_i + c_i \geq 1$. Since $D_0 + \rho \leq q - x + T_X + h < 3X$ for large X , there are at most Λ_X of them. Selecting one representative for each distinct r_i leaves at least $\lfloor \ell/\Lambda_X^2 \rfloor$ slab points: if $M \geq \lfloor \ell/\Lambda_X \rfloor$ matched edges remain and at most Λ_X have the same r_i , then $\lceil M/\Lambda_X \rceil \geq \lfloor \ell/\Lambda_X^2 \rfloor$. Indeed, writing $\ell = a\Lambda_X^2 + r$ with $0 \leq r < \Lambda_X^2$, one has $\lfloor \ell/\Lambda_X \rfloor \geq a\Lambda_X$, and hence $\lceil \lfloor \ell/\Lambda_X \rfloor / \Lambda_X \rceil \geq a = \lfloor \ell/\Lambda_X^2 \rfloor$. The retained points have distinct labels and the stated divisibilities. $\square$

7 The Shifted-CRT Slab Count

Proposition 7.4 is the only counting statement needed after terminal support has been extracted. Its proof has three parts: an auxiliary interpolation step, a nonlinear component count using Input 2.3, and a separate estimate for line components.

Lemma 7.1 (Auxiliary interpolation). *Let Z be a set of $K \geq 1$ distinct integer points in a fixed polylogarithmic box*

$$|A|, |R| \leq (\log X)^{B_0},$$

and assume $K \log \log X = o(\log X)$. With

$$m = \left\lceil 4\sqrt{K} \right\rceil,$$

there is a nonzero nonconstant polynomial $G(A, R) \in \mathbb{Z}[A, R]$, of total degree at most m and height $\exp(O(m \log \log X))$, vanishing on Z . It may be chosen primitive, with a fixed sign convention for its first nonzero coefficient.

Proof. The number N_m of monomials $A^u R^v$, $u + v \leq m$, is $(m+1)(m+2)/2 > 2K$. The $K \times N_m$ evaluation matrix on Z has all entries bounded by $\exp(O(m \log \log X))$. Compare coefficient vectors in $[0, Q)^{N_m}$ under the K evaluation forms. For

$$Q \geq (3N_m H_0)^{K/(N_m-K)} + 1, \quad H_0 = \exp(O(m \log \log X)),$$

two vectors have the same image; their nonzero difference gives an integer kernel vector of height $\exp(O(m \log \log X))$. The corresponding polynomial vanishes on Z , and is not a nonzero constant because $K \geq 1$. Dividing its coefficient vector by the gcd of the coefficients and fixing the sign of the first nonzero coefficient preserves all these bounds and gives the primitive normalized choice. $\square$

Lemma 7.2 (Line-factor height). *Let $G(A, R) \in \mathbb{Z}[A, R]$ be primitive, nonzero, of total degree at most m , with*

$$m^2 = o(\log X), \quad H(G) = \exp(o(\log X)).$$

If

$$L(A, R) = \alpha A + \beta R + \gamma$$

is a primitive integer line factor of G over $\mathbb{Q}$, then

$$\max(|\alpha|, |\beta|, |\gamma|) = X^{o(1)}.$$

Proof. Homogenize G as a degree- m form; the line becomes the primitive homogeneous factor

$$L^h = \alpha A + \beta R + \gamma W.$$

Permute variables so the largest coefficient is the coefficient c of the homogenizing variable, and write the line as $aU + bV + cW$. Dehomogenizing at $W = 1$ gives a polynomial $P(U, V)$ of degree at most m , height $\leq H(G)$, divisible by $aU + bV + c$.

If $\max(|a|, |b|, |c|) = 1$ there is nothing to prove. Otherwise choose an integer slope $0 \leq t \leq m + 1$ such that $a + bt \neq 0$ and the cofactor in

$$P = (aU + bV + c)Q$$

does not vanish identically on $(U, V) = (Z, tZ)$. In this case $(a, b) \neq (0, 0)$. Such a slope exists because at most $m - 1$ slopes kill the nonzero cofactor and at most one slope makes $a + bt = 0$: writing $Q = \sum_j Q_j$ into homogeneous parts, if $Q(Z, tZ)$ is identically zero, then $Q_j(1, t) = 0$ for every j , and one fixed nonzero Q_j has degree at most $m - 1$. Then

$$P_t(Z) = P(Z, tZ)$$

is a nonzero one-variable integer polynomial of degree at most m , divisible by $(a + bt)Z + c$, and

$$\|P_t\|_2 \leq \exp(O(m \log m))H(G).$$

Mignotte's inequality, Input 2.4, gives

$$\max(|a + bt|, |c|) \leq 2^m \|P_t\|_2 \leq \exp(O(m \log m))H(G).$$

Since $|c|$ was the largest line coefficient and $m \log m = O(m^2) = o(\log X)$, the result follows. $\square$

Lemma 7.3 (Nonabsolute factors). *Let $H(A, R) \in \mathbb{Z}[A, R]$ be primitive, squarefree, $\mathbb{Q}$ -irreducible, not absolutely irreducible, and of total degree $d \geq 2$. Then*

$$\#\{P \in \mathbb{Q}^2 : H(P) = 0\} \leq d^2.$$

Proof. Over $\overline{\mathbb{Q}}$, the polynomial H is a product of at least two distinct Galois-conjugate absolute factors. A rational zero of H lies on one absolute factor, hence by Galois conjugacy lies on every conjugate factor. Choose two distinct factors. They have no common component, so Bezout bounds their common affine zero set by the product of their degrees, at most d^2 . $\square$

Proposition 7.4 (Shifted-CRT slab count). *Fix constants $A, B, B_1, \kappa > 0$. In a dyadic shell $X < n \leq 2X$, suppose the endpoints under consideration have integral localized data*

$$n = q + h, \quad 1 \leq h \leq (\log X)^A, \quad T_X = (\log X)^B,$$

$$x = q^\theta + O(T_X), \quad D_0 = q - x > 0, \quad 0 < \theta < 1,$$

where the implicit constant in $O(T_X)$ is uniform over all endpoints with the fixed parameter tuple.

$$(\log X)^{-B} \leq 1 - \theta \leq B \frac{\log \log X}{\log X}.$$

Assume that, for every fixed $\varepsilon > 0$, the number of parameter tuples, including h, θ , and K , is $O_\varepsilon(X^\varepsilon)$. Let

$$Y_X = \exp(\log X / (\log \log X)^\kappa).$$

Suppose each endpoint carries K distinct local slab points

$$z_i = (a_i, r_i), \quad 0 \leq a_i < T_X, \quad |r_i| \leq (\log X)^{B_1},$$

with distinct prime labels $p_i > Y_X$ satisfying

$$p_i \mid x - a_i, \quad p_i \mid D_0 + r_i.$$

If

$$K \log \log X = o(\log X), \quad \frac{K \log Y_X}{\sqrt{K} \log X} \rightarrow \infty,$$

then the number of such endpoints in $(X, 2X]$ is $o(X)$.

Proof. We divide the proof into the three counting mechanisms described above.

Auxiliary curves. Fix one localized parameter tuple and one retained K -set Z of local points. The number of possible Z 's is $X^{o(1)}$, since the local box is polylogarithmic and

$$\binom{(\log X)^{O(1)}}{K} \leq \exp(O(K \log \log X)) = X^{o(1)}.$$

Fix a constant C_G , depending only on the local box exponents B and B_1 , for which Lemma 7.1 supplies a primitive nonzero polynomial $G(A, R)$ of degree at most $m = \lceil 4\sqrt{K} \rceil$, height at most $\exp(C_G m \log \log X)$, and vanishing on Z . Among all such polynomials with first nonzero coefficient positive, choose the lexicographically first coefficient vector. Thus G depends only on the fixed parameters and the coordinate set Z , not on the endpoint-specific prime labels.

For every endpoint and every retained point,

$$(x, -D_0) \equiv (a_i, r_i) \pmod{p_i}.$$

Therefore each p_i divides $G(x, -D_0)$, and the distinctness of the labels gives

$$\prod_i p_i \mid G(x, -D_0).$$

If $G(x, -D_0) \neq 0$, then, since G has $O(m^2) = O(K)$ monomials and $|x|, |D_0| = O(X)$,

$$\log |G(x, -D_0)| = O(m \log X) = O(\sqrt{K} \log X).$$

But

$$\log \prod_i p_i \geq K \log Y_X,$$

and the ratio $K \log Y_X / (\sqrt{K} \log X)$ tends to infinity. This is incompatible with the preceding bound for large X . Hence $G(x, -D_0) = 0$.

Nonlinear components. Thus all endpoints lie on $X^{o(1)}$ auxiliary curves $G(A, R) = 0$, each of degree $m = X^{o(1)}$. Remove content and repeated factors, and then consider the primitive squarefree factorization over $\mathbb{Q}$. This does not change the zero set relevant to us: every point with $G(x, -D_0) = 0$ lies on at least one primitive squarefree $\mathbb{Q}$ -irreducible factor. Primitive line factors are deferred to the next paragraph. Nonlinear absolutely irreducible factors of degree $d \geq 2$ are integral affine curves. For one such factor, Input 2.3, in the box of height $O(X)$, gives

$$O(d^3 X^{1/d} (\log X + d)) \leq X^{1/2+o(1)}$$

points. The constant in Input 2.3 depends only on the fixed ambient dimension, and the squarefree factors of one auxiliary curve have total degree at most $m = X^{o(1)}$; summing this bound over the absolutely irreducible nonlinear factors still gives $X^{1/2+o(1)}$ points for that auxiliary curve. If a $\mathbb{Q}$ -irreducible nonlinear factor is not absolutely irreducible, Lemma 7.3 gives $O(d^2)$ rational points, so all nonabsolute factors of one auxiliary curve contribute only $X^{o(1)}$ points. Multiplication by the $X^{o(1)}$ curves still gives $o(X)$ endpoints, since the fixed parameter tuple includes h , and $(x, -D_0)$ determines $q = x + D_0$ and $n = q + h$.

Line components. It remains to count primitive line factors

$$\alpha A + \beta R + \gamma = 0.$$

Every such factor divides the original primitive auxiliary polynomial G over $\mathbb{Q}$, even after passing to the squarefree radical, so Lemma 7.2 applies to the same G . By Lemma 7.2,

$$M_X := \max(|\alpha|, |\beta|, |\gamma|) = X^{o(1)}$$

uniformly for all such factors. At the global point $R = -D_0 = x - q$,

$$(\alpha + \beta)x - \beta q + \gamma = 0. \quad (1)$$

Also $q \geq X/2$. Since $x = q^\theta + O(T_X)$ and

$$q^\theta \geq q/(\log X)^{O(B)} \gg T_X,$$

we have

$$x \gg q/(\log X)^{O(B)}.$$

For large X , neither β nor $\alpha + \beta$ can be zero in (1), otherwise x or q would be $X^{o(1)}$.

Define

$$F(q) = (\alpha + \beta)q^\theta - \beta q + \gamma.$$

Since $x = q^\theta + O(T_X)$, equation (1) gives, for a fixed constant C_1 ,

$$|F(q)| \leq C_1 M_X T_X. \quad (2)$$

On the sublevel set (2), we have

$$|F'(q)| \geq c|\beta|(1 - \theta). \quad (3)$$

Indeed, if $F'(q) = \delta$ with $|\delta| < |\beta|(1 - \theta)/4$, then

$$(\alpha + \beta)q^\theta = \frac{(\beta + \delta)q}{\theta}$$

and hence

$$F(q) = \frac{\beta q(1 - \theta)}{\theta} + \frac{\delta q}{\theta} + \gamma.$$

Using $|\beta| \geq 1$, $q \gg X$, $(1 - \theta) \geq (\log X)^{-B}$, and $\theta \geq 1/2$, the first term in this last display has magnitude $\gg X/(\log X)^B$. The δ -term has at most one quarter of that magnitude, and $|\gamma| \leq M_X = X^{o(1)}$ is negligible in comparison. Hence $|F(q)| \gg X/(\log X)^B$, which is much larger than $C_1 M_X T_X$. This contradicts (2), proving (3).

The derivative F' is monotone on $q > 0$, since $0 < \theta < 1$ and $\alpha + \beta \neq 0$, so F' has at most one zero. Thus F is monotone on at most two intervals,

and the sublevel set $|F(q)| \leq C_1 M_X T_X$ has at most two interval components. Each component has length

$$O\left(\frac{C_1 M_X T_X}{|\beta|(1-\theta)}\right) = X^{o(1)}$$

by the mean-value theorem and (3). Hence each line factor contributes $X^{o(1)}$ integer values of q , and, since h is fixed by the parameter tuple, $X^{o(1)}$ endpoints $n = q + h$. There are only $X^{o(1)}$ curves and line factors. Thus all line components contribute $o(X)$. Combining the nonlinear and line estimates proves the proposition. $\square$

8 Counting Nonempty Rejections

We now combine the preceding modules. All exceptional families already have shell size $o(X)$; every endpoint that remains supplies enough terminal support to fall under Proposition 7.4.

Proposition 8.1. *For the modified greedy construction using the deterministic deletion set from Proposition 4.1, the number of nonempty rejections $n \leq N$ is $o(N)$.*

Proof. Work in a dyadic shell $X < n \leq 2X$. Remove the deterministic deletions, empty-suffix endpoints, and the exceptional families from Propositions 4.1 and 5.1. These contribute $o(X)$ in the shell: the deterministic deletions are density zero, the empty-suffix endpoints are controlled by Corollary 3.6 with ambient bound $2X$, and the localization exceptions were proved either as shell $o(X)$ families or as global $o(N)$ families applied with $N = 2X$. These estimates were established independently of the present proposition.

For every remaining endpoint, Proposition 5.1 applies. Discard the $o(X)$ smooth-left family of Lemma 6.1. Then every left factor has a chosen label $p_i = P^+(x - a_i) > Y_X$, where

$$Y_X = \exp(\log X / (\log \log X)^2).$$

By Lemma 6.2, the endpoint supplies at least $\lfloor \ell / \Lambda_X^2 \rfloor$ slab points, where

$$\Lambda_X = \left\lceil \frac{\log(3X)}{\log Y_X} \right\rceil + 2 = (\log \log X)^2 + O(1).$$

Since Proposition 5.1 gives

$$\ell \geq b \geq c \frac{\log X}{\log \log X},$$

we may retain the lexicographically first

$$K_{\text{slab}} = \left\lfloor \frac{c \log X}{100(\log \log X)^5} \right\rfloor$$

of these extracted slab points. The factor 100 leaves room for the floor loss: for large X , $\Lambda_X \leq 2(\log \log X)^2$, so

$$\frac{\ell}{\Lambda_X^2} \geq \frac{c \log X}{4(\log \log X)^5},$$

and hence

$$\frac{\ell}{\Lambda_X^2} - 1 \geq \frac{c \log X}{100(\log \log X)^5} \geq K_{\text{slab}}.$$

Thus $\lfloor \ell/\Lambda_X^2 \rfloor \geq K_{\text{slab}}$.

The hypotheses of Proposition 7.4 hold with $A = 7$, $\kappa = 2$, $B_1 = B + 1$, $K = K_{\text{slab}}$, and $\theta = b/\ell$. Indeed, Proposition 5.1 gives $n = q + h$, $h \leq (\log X)^7$, $x = q^\theta + O(T_X)$ with a fixed implicit constant, $D_0 = q - x > 0$, $0 < \theta < 1$, and the two required bounds on $1 - \theta$. Lemma 6.2 gives the distinct slab points and distinct prime labels with the two divisibilities; retaining the first K_{slab} points preserves these properties. Finally,

$$|r_i| = |a_i + c_i| \leq T_X + (\log X)^7 \leq (\log X)^{B+1}$$

for large X . The localized parameter tuple is determined by h , (b, ℓ) , and K_{slab} . Since $h \leq (\log X)^7$ and, by Proposition 5.1,

$$1 \leq b, \ell \leq \lceil ((\log X)^7 + 1) \log_2(2X) \rceil + 2 = O((\log X)^8),$$

this gives only $O((\log X)^{23}) = X^{o(1)}$ possibilities. Moreover

$$K_{\text{slab}} \log \log X = o(\log X),$$

and, since

$$K_{\text{slab}} \geq \frac{c \log X}{200(\log \log X)^5}$$

for all large X ,

$$\frac{K_{\text{slab}} \log Y_X}{\sqrt{K_{\text{slab}}} \log X} = \frac{\sqrt{K_{\text{slab}}}}{(\log \log X)^2} \rightarrow \infty.$$

Thus Proposition 7.4 counts the surviving endpoints by $o(X)$. Adding back the finitely many $o(X)$ families removed above, every dyadic shell contains $o(X)$ nonempty rejections.

Finally sum dyadic shells. If f_j is the ratio of the number of nonempty rejections in $(2^j, 2^{j+1}]$ to 2^j , then $f_j \rightarrow 0$. For every $\varepsilon > 0$, choose J_0 so that $f_j \leq \varepsilon$ for $j \geq J_0$. The finite initial shells contribute $O(2^{J_0})$, while the dyadic tail up to N contributes at most

$$\varepsilon \sum_{2^j \leq N} 2^j \leq 2\varepsilon N.$$

Letting $\varepsilon \rightarrow 0$ gives $o(N)$. □

9 Proof of the Main Theorem

Proof of Theorem 1.1. Use the deterministic deletion set D from Proposition 4.1 and run the modified greedy construction. By Proposition 4.1, forced deletions contribute $o(N)$ rejected values. By Corollary 3.6, empty-suffix collision rejections contribute $o(N)$. By Proposition 8.1, nonempty collision rejections contribute $o(N)$.

Every rejected n belongs to the union of these three families. If $n \in D$, it is forced. If $n \notin D$, Lemma 3.3 gives an old-new witness $P = nS$. If some witness has $S = 1$, n is empty-suffix; otherwise the deterministic chosen witness has $S > 1$, and n is counted by the nonempty family. Disjointness is unnecessary. Hence the total rejected set has density zero, so the accepted set has density one. Lemma 3.2 gives distinct consecutive-block products at every finite stage. The increasing enumeration of the accepted set proves the theorem. □

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